MILITARY SPECIFICATION

POWER SUPPLY PP-4763( )/GRC

1. SCOPE

1.1 Scope. - This document covers a power supply designated PP-4763( )/GRC rated for a nominal 28 volt dc, 50 ampere continuous output, suitable for operating an AN/GRC-106 Radio Set. (See 6.1 and 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-781	Steel Strapping, Flat
TT-E-529	Enamel, Alkyd, Semi-gloss
TT-P-636	Primer coating, Synthetic, Wood and Ferrous Metal
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lockcorner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) Cut Shapes
PPP-T-97	Tape Pressure sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-131	Barrier Material, Water vapor proof, Flexible
MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment

FSC 5820

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

MIL-V-173	Varnish, Moisture-and-Fungus-resistant (For The Treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-S-901	Shock Tests, H.I. (High Impact), Shipboard Machinery, Equipment and Systems, Requirements for (Navy)
MIL-C-5541	Chemicals Films and Chemical Film Materials For Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-P-11268	Part, Materials and Processes Used in Electronic Communications Equipment
MIL-M-13231	Marking of Electronic Items
MIL-P-13949	Plastic Sheet, Laminated, Copper Clad (For Printed Wiring)
MIL-P-15328	Primer (Wash), Pretreatment, Blue (Formula No. 117-B For Metals)
MIL-P-18177	Plastic Sheet, Laminated, Thermosetting, Glass Fiber Base, Epoxy-resin

STANDARDS

FEDERAL

Fed-STD-595	Color

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 inch x 48 inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-785	Requirements for Reliability Program
MIL-STD-810	Environmental Test Methods

DRAWINGS (ECOM)

DL-SM-A-640000	Power Supply PP-4763( )/GRC

(Copies of specifications, standards, and drawings, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipuated when requesting copies).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description. - The power supply is a Solid State unit rated for a nominal 28 volt d-c and 50 ampere continuous output, with an input of 115 or 230 volt, 50-60 Hz. In addition to meeting all of the detailed specifications below, the unit is suitable for operating an AN/GRC-106 radio set when used in shelter configurations such as the AN/GRC-122 and the AN/GRC-142.

3.2 Preproduction samples. - The contractor shall furnish three (3) Preproduction samples for approval if required by the invitation for bids and contract. (See 4.3 and 6.2d(1))

3.3 Construction. - The equipment shall be constructed in accordance with Drawing DL-SM-A-640000.

3.4 Parts, materials, and processes; general. - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply. (See 6.6)

3.5 Finish. - All cabinet parts of the unit shall be finished as follows:

3.5.1 Steel parts. - All external steel parts shall be cadmium plated per QQ-P-416 Type II, Class 2. All mating surfaces of the external enclosure shall remain unpainted for RFI bonding. Painted surfaces shall be primed with sandwich prime zinc chromate per MIL-P-8585 and shall be finished with synthetic enamel per TT-E-529.

3.5.1.1 Interior steel parts not requiring cadmium plating shall be primed with iron oxide per TT-P-636 and shall be finished with synthetic enamel per TT-E-529.

3.5.2 Aluminum parts. - All external aluminum parts shall be plated with alodine 1200 per MIL-C-5541. All exterior mating surfaces shall be left unpainted for RFI bonding. Painted surfaces will be primed with sandwich prime zinc chromate per MIL-P-8585 and will be finished with synthetic enamel per TT-E-529.

3.5.2.1 Interior unplated parts will receive a wash primer pre-treatment per MIL-C-15328, then primed with sandwich prime zinc chromate per MIL-P-8585 and finished with synthetic enamel per TT-E-529.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.5.3 Color. - All painted parts shall be olive drab color No. 24087 of Fed. Std. 595.

3.6 Marking. -

3.6.1 General. - Marking shall conform to Specification MIL-M-13231. Front panel marking shall be group I as described in that specification. (See 4.4)

3.6.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.6.3 Serial numbers. - Each Power Supply PP-4763( )/GRC shall be serial numbered.

3.7 Moisture and Fungus proof (MFP) treatment for equipment and assemblies. - When overall tropicalization treatment of equipment is required to enable the equipment to comply with the fungus requirement of 3.7.2, a moisture and fungus resistant coating of varnish conforming to MIL-V-173 shall be applied in accordance with MIL-T-152. The moisture-fungus resistant coating shall be applied after the equipment has been cleaned to remove such contaminants as corrosion products, dust, fingerprints, lubricating oils, mold release agents, sand, and solder fluxes. When tape is used on parts of equipment that are masked against treatments, the tape shall be an electrical grade. In the application of the MFP coating, care shall be taken to assure that the coating does not affect circuit constants and thereby degrade performance.

3.7.1 Statement of Treatment. - The statement of treatment shall indicate whether or not nutrient materials have been employed. If nutrient materials have been used, it shall also describe in detail the materials to be treated and the treating compounds and processes that the contractor proposes to use.

3.7.2 Fungus. - The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.8.9. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish of metal surface.

3.8 Printed circuit boards. - (See 4.4) The material for printed circuit boards shall be type FR, per MIL-P-18177 and MIL-P-13949. It is required that printed circuit boards, when used, receive the following conformal coating and DO NOT receive any MIL-V-173 MFP coating at anytime.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.8.1 Coating. - The completed printed wiring assembly shall be covered with a non-corrosive, moisture and fungus resistant coating. The coating used shall be tough, flexible polyurethane resin. It shall provide electrical insulation, abrasion resistance, adhesion to metals and nonmetals, and protection against environmental conditions. The coating resin shall not be applied to any part or surface where such application will interfere with the operations or performance of the equipment. Upon examination, the cured coating shall present a smooth, continuous surface. The coating shall pass from the mounted components to the circuit board with no break at their junction. Except for the areas masked, all surfaces shall be covered with a coating of .001 inch minimum to .005 maximum thickness. One, two, three, and five watt wire wound resistors shall not be coated.

3.8.2 Cured coating. - The cured coating shall meet the following requirements:

(a) Coating shall be free of visible dust particles or other foreign matter.

(b) It shall be smooth and uniform and have no globules or bare spots.

(c) It shall be substantially free of bubbles.

(d) There shall be no blisters or cracks.

(e) It shall be transparent and shall not affect marking legibility.

(f) It shall be capable of being removed in local areas for replacement of components by a soldering iron.

(g) When applied to a test board of 1/16 inch thickness, the coating shall not rupture or separate from the board when the board is bent into an arc of twelve inches radius.

3.9 Safety. -

3.9.1 Capacitor discharge. - Adequate means shall be provided to discharge all capacitors on the input and output circuits when the unit is de-energized, regardless of the setting of the primary circuit breaker(s).

3.9.2 Interlock switches. - Interlock switches will be provided to turn input power off when the front panel is opened or the top is removed.

3.9.3 Warning decals. - A warning notice shall be provided on the inside of the hinged front panel to indicate the presence of 115 to 230 VAC on the terminals of the main circuit breakers. Warning decals shall also indicate other points of high voltage as, for example, the input A.C. circuit terminals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.9.4 Circuit breaker cover. - An insulated cover shall be provided for the circuit breaker terminals.

3.10 Test Points. - Test points shall be provided and identified for all voltage points on the main chassis and also on the printed circuit control board.

3.11 Electrical. -

3.11.1 Input voltage. (See 4.7.1) - The power supply shall deliver the output required by this specification (see 3.11.2) from the following single phase input sources:

Volts AC (± 10%)	Frequency (Hz) (± 5%)
115	50, 60
230	50, 60

3.11.2 DC output current. (See 4.7.1) - The power supply shall be capable of delivering 50 amperes continuously at 29 volts d.c. at all rated input voltages and frequencies under normal and environmental conditions specified herein.

3.11.3 DC output voltage. (See 4.7.1) - The power supply shall have a nominal d.c. output voltage of 28 volts with an adjustment range of 27 to 29 volts minimum (22 to 30 volts maximum) by means of a screwdriver adjustment behind a small access plate on the front panel. The unit shall not deliver more than 30 volts d.c. steady state.

3.11.4 Static voltage regulation. (See 4.7.2) - The power supply shall maintain a d.c. volt age regulation at its output terminals of ± 0.5% against any combination of load currents and line voltages within the rated ranges. The regulation shall be accomplished by means of a phase controlled SCR regulator.

3.11.5 Dynamic voltage regulation. (See 4.7.3) - The power supply shall not produce any transients exceeding 5 volts at its output terminals as a result of step load changes from no-load to full-load or full-load to no-load or ± 10% step changes in a.c. line voltage. The d.c. voltage shall not overshoot more than 5 volts over the set voltage upon a.c. turn-on or turn-off. The response time (63% of transient) of the unit shall not exceed 75 milliseconds upon application of full load or 125 milliseconds upon removal of full load.

3.11.6 Ripple voltage. - (See 4.7.4) - The output ripple voltage of the power supply shall not exceed 1% rms under rated line and load conditions.

3.11.7 Insulation resistance. (See 4.7.5). - The resistance from primary to secondary or from any winding to the core of the input transformer shall not be less than five megohms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.12 Interchangeability. - Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.9)

3.13 Equipment Protection. (See 4.7.6)

3.13.1 Overcurrent. (See 4.7.6.1) - The power supply shall be protected against heavy overloads and short circuits by means of circuit breakers in the a.c. input line.

3.13.2 Overvoltage. (See 4.7.6.2) - The power supply shall be provided with a clamping circuit to prevent the output voltage from rising above 35 volts and to subsequently shut the unit down under fault conditions.

3.14 Service conditions. -- The equipment shall meet the following service conditions:

3.14.1 Operation (general). - The equipment shall operate under environmental service operating conditions specified in 3.14.2 through 3.14.2.7 with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement and 6.6.

3.14.2 Service conditions operation requirements. - Prior to, during, and after testing as specified for each of the following service conditions, the equipment shall meet the requirements of 3.11.1 and 3.11.2.

3.14.2.1 Low Temperature. - See 4.8.1.

3.14.2.2 High Temperature. - See 4.8.2.

3.14.2.3 Humidity (moisture-resistance). - See 4.8.3.

3.14.2.4 Dust. - See 4.8.4.

3.14.2.5 Salt Fog. - See 4.8.5.

3.14.2.6 Vibration. - See 4.8.6.

3.14.2.7 Shock. (See 4.8.7). - After being subjected to the shock test of MIL-S-901 the power supply shall meet the output requirements of 3.11.2 and any physical damage shall be minor only. A trip off of the circuit breaker during any blow in the series of shock tests is permissible. The circuit breaker may be manually reset to the "on" position after any blow, if trip-off occurs.

3.15 Radio frequency interference. - The radio frequency interference limits shall be in accordance with MIL-I-6181. (See 4.8.8)

3.16 Preconditioning. - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6. (also see 4.5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.17 Reliability. - The reliability of the PP-4763 power supply equipments shall provide a mean-time-between-failures (MTBF) of 7500 hours. The contractor shall conduct the following reliability tasks in accordance with MIL-STD-781 and MIL-STD-785:

Periodic Reports
Reliability Demonstration
Failure Data Collection & Analysis
Test Plans

The reliability of PP-4763 equipments shall be demonstrated in accordance with paragraph 4.12 of this specification.

3.18 Technical literature, tools, and running spare parts. - Technical literature, tools and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2(g))

3.19 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirements 9 of MIL-STD-454, Specification MIL-P-11268, and following paragraphs of this document:

3.9 Safety (see 4.10)

3.10 Test points (see 4.10)

4. Quality Assurance Provisions

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.11)

4.3 Preproduction inspection. - This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, and group C (See Tables II and III respectively). The environmental preproduction inspection shall be performed in the order shown in table I.

Table I. - Order of Preproduction inspection

Inspection (For additional preproduction inspection see 4.3)	Req. Para.	Insp. Para.
Sequence of inspection when three units are available		
Unit 1		
Low Temperature	3.14.2.1	4.8.1
High Temperature	3.14.2.2	4.8.2
Radio frequency interference	3.15	4.8.8
Humidity	3.14.2.3	4.8.3
Unit 2		
Salt fog	3.14.2.5	4.8.5
Unit 3		
Dust	3.14.2.4	4.8.4
Vibration	3.14.2.6	4.8.6
Shock	3.14.2.7	4.8.7
Fungus	3.7.2	4.8.9

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery.

Item	Where required
Finish	3.5
Marking	3.6
Printed circuit boards	3.8

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection of equipment before preparation for delivery. - Each unit of each lot of equipment shall be subjected to bounce preconditioning (see 3.16) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion-- may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A Inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table II. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105, using the general inspection levels and the AQL's indicated in Table II. Group A inspection shall be performed in any order which is satisfactory to the Government except that the Insulation Resistance test shall be performed first prior to performing any electrical tests.

Table II - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL	
			Major	Minor
Visual and mechanical	3.19	4.10	1.0%	4.0%
Electrical				
Insulation Resistance	3.11.7	4.7.5		
AC input	3.11.1	4.7.1	1.5	
DC output current	3.11.2	4.7.1	dphu	
DC output voltage	3.11.3	4.7.1	for the	
Static voltage regulation	3.11.4	4.7.2	group	
Dynamic voltage regulation	3.11.5	4.7.3	combined	
Ripple voltage	3.11.6	4.7.4		

NOTE: All electrical defects considered Major.

4.5.2 Group B inspection. - This inspection including sampling shall consist of the inspection for interchangeability and equipment Protection, paragraphs 3.12 and 3.13 respectively and shall conform to special procedures for small sample inspection of MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be S-4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection. The inspection for interchangeability (See 3.12) will not require stripping units of any components.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.5.3 Group C inspection. - This inspection shall consist of the tests specified in Table III and shall be performed on sample units that have been subjected to and met Groups A and B inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1 and shall be subjected to the tests in the order listed.

Table III - Group C inspection

Inspection	Req. Para.	Insp. Para.
Group I		
Low Temperature	3.14.2.1	4.8.1
High Temperature	3.14.2.2	4.8.2
Humidity	3.14.2.3	4.8.3
Group II		
Salt fog	3.14.2.5	4.8.5
Group III		
Dust	3.14.2.4	4.8.4
Vibration	3.14.2.6	4.8.6
Shock	3.14.2.7	4.8.7
Fungus	3.7.2	4.8.9
Group IV		
Radio frequency interference (Not required when conducted on First Article and test report is included with the preproduction test report, unless otherwise specified.	3.15	4.8.8

4.5.3.1 Group C sampling. - For the initial group C tests, one sample for each group (4 samples total) shall be selected at random from the first 25 units produced and subjected to the inspections listed in Table III. Thereafter, one sample each for Groups I, II, III (3 samples total) shall be selected from each 250 units, or fraction thereof produced and subjected to the inspections for those groups as listed in Table III. No subsequent inspection for the test in Group IV will be required.

4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable, and (2) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the currentversion before use.

4.5.3.2 (Cont'd) Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective.

4.5.3.3 Reinspection of conforming group B and C sample units. - Unless otherwise specified, sample units which have been subjected to and passed group B and group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Bounce preconditioning. - The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneatales, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of three (3) inches as the unit bounces. The unit shall be subjected to this preconditioning for one (1) minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to inspection. (see 3.16)

4.7 Electrical tests. -

4.7.1 AC Input and DC output. - The output voltage shall meet the requirement of 3.11.1, 3.11.2 and 3.11.3 full load, when the input voltage and frequency is varied as specified in Table IV:

Table IV - Input test conditions

Input Voltage (A.C.)	Freq. (HZ)
103.5	47.5, 63
115	60
126.5	47.5, 63
207	47.5, 63
230	60
253	47.5, 63

4.7.2 Static voltage regulation (See 3.11.4). - Using the input test conditions specified in Table IV, Power Supply PP-4763/GRC shall meet the d.c. voltage regulation requirements of 3.11.4. The load shall be resistive and regulation shall be checked at 0, 25 and 50 amp load conditions. The percent regulation shall be determined by the following formula:

$$\% \text{ Regulation} = \frac{(\text{Maximum Voltage} - \text{Minimum Voltage})}{\text{Nominal Voltage}} \times 100$$

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:18Z
Check the source to verify that this is the currentversion before use.

4.7.3 Dynamic voltage regulation (See 3.11.5). - Using each input test condition specified in Table IV the following procedure shall be followed to determine compliance with the requirements of 3.11.5.

(a) Operate the power supply at 28VDC (± 0.1V) output with no load connected. At initial turn-on, observe the output d.c. voltage for overshoot over the set voltage. Turn-off the equipment, and again observe the output d.c. voltage for overshoot.

(b) Reapply the a.c. input power, apply a 50 ampere load (0.56 ohm resistance) by a step change (sudden closing of a knife switch or high current relay) and observe the voltage response on an oscilloscope accurately calibrated as to time base and vertical voltage.

(c) Remove the a.c. input voltage and observe the output d.c. voltage for overshoot of the set voltage observed in pa. (b). Reapply the AC input power and again observe for overshoot.

(d) Remove the 50 ampere load by suddenly opening the output circuit and observe the voltage response on the oscilloscope.

(e) At the 115 and 230VAC input test points, full load, the input voltage shall be suddenly varied plus 10% and minus 10% from the nominal. During the variation manipulation, the d.c. output shall be observed for overshoot.

4.7.4 Ripple voltage (see 3.11.6). - Using the input test conditions specified in Table IV, the ripple voltage at full load shall be measured with an electronic voltmeter. The percent of ripple shall be determined by the following formula:

$$\% \text{ Voltage ripple} = \frac{\text{A.C. ripple voltage (RMS)}}{\text{D.C. output voltage}} \times 100$$

4.7.5 Insulation Resistance (See 3.11.7). - The following procedure shall be followed to determine compliance with 3.11.7.

(a) Remove Plug-in module P/C 101.

(b) Short out all semiconductors.

(c) Using jumper leads, connect the secondary leads from T1 together and to ground.

(d) Connect the primary leads from T1 together.

(e) Connect the insulation tester (500VDC, 0-200 megohm) between T1 primary and ground. Operate the Insulation Tester for one minute and observe the reading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

(f) Remove the jumper lead from T1 secondary and ground.

(g) Connect the insulation tester between secondary leads and ground and operate the tester for one minute. Observe the reading.

(h) Disconnect the insulation tester, remove all jumper leads and shorts, and return the test unit to its normal operating condition.

4.7.6 Equipment Protection (See 3.13)

4.7.6.1 Overcurrent (See 3.13.1). - The power supply shall be operated from a single phase 115 VAC, 60 HZ power source. A variable load shall be connected across the test item's output terminals and the load shall be increased until the main circuit breaker is activated, thus shutting off the test item. The input current required to activate the main circuit breaker shall not exceed thirty (30) amps. The power supply shall then be operated at nominal requirements (28 VDC, 50 amps). The output terminals shall be short-circuited; thus shutting off the test item. The short shall then be removed and the test item operated for five (5) minutes. The proceeding test shall be repeated using a 230 VAC, 60 HZ power input with the input current required to activate the main circuit breaker not to exceed fifteen (15) amps.

4.7.6.2 Output Overvoltage Test (See 3.13.2)

(a) This test is performed with no output load.
(b) Connect a jumper lead (#8 wire or larger) across SCR-1.
(c) Increase the a.c. line voltage from 0 volts until the power supply breaker trips off.
(d) Record supply output voltage at point of breaker trip off. This voltage shall not be less than 34 or greater than 35 volts.

4.8 Temperature test.

4.8.1 Low Temperature (See 3.14.2.1). - The equipment shall be tested in accordance with Method 502, Procedure 1, of MIL-STD-810. The storage temperature shall be - 80°F and the operating temperature shall be - 40°F. Exposure at the low storage temperature extreme shall be maintained until temperature stabilization of the test item is reached, but not to exceed 24 hours. Exposure at the low operating temperature extreme not to exceed 72 hours. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.14.2.

4.8.2 High Temperature (See 3.14.2.2). - The equipment shall be tested in accordance with Method 501, Procedure 2 of MIL-STD-810. The following additions and modifications apply and will govern during the High Temperature tests.

Step 6: Repeat steps 3, 4 & 5 one additional time. (Making a total of two (2) twelve (12) hour cycles)

Step 7: Operating temperature is + 55°C (+ 131°F)

Step 8: Operate test item for a period of 8 hours using a 50 ampere output load.

The equipment shall meet specification performance for the measurements specified in 3.14.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.8.3 Humidity (moisture-resistance) test. - (See 3.14.2.3). - The equipment shall be tested in accordance with Method 507, Procedure 1, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.14.2. After completion of the test, the power supply shall be connected to a single phase, 230 VAC, 60 Hz power source and operated under full output load conditions (28VDC at 50 amperes) for a period of not less than ten minutes. Examine for evidence of damage or deterioration. There shall be no deterioration, corrosion, or change in tolerance limits of any internal or external components which could in any manner prevent the unit from meeting operational requirements.

4.8.4 Dust Test-(See 3.14.2.4). - The power supply shall be subjected to the test as specified in Method 510 of MIL-STD-810. Step 3 of Method 510 may be performed immediately after reaching stabilization in Step 2. After completion of the test, the power supply shall be connected to a single phase, 230 VAC, 60 Hz power source and operated under full output load conditions (28 VDC at 50 amperes) for a period of not less than ten (10) minutes. The test item shall also be examined for evidence of damage or deterioration.

4.8.5 Salt Fog test (See 3.14.2.5). - After completion of the salt fog test in accordance with Method 509, Procedure 1, of MIL-STD-810, the power supply shall be connected to a single phase, 230 VAC, 60 Hz power source and operated under full output load conditions (28 VDC at 50 amperes) for a period of not less than ten (10) minutes. The test item shall also be examined for evidence of damage or deterioation.

4.8.6 Vibration (See 3.14.2.6). - After being subjected to the test specified in MIL-STD-167 (Type I equipment) the power supply shall be connected to a single phase, 230 VAC, 60 Hz power source and operated under full output load conditions (28 VDC at 50 amperes) for a period of not less than ten (10) minutes. In addition, the test specified in paragraph 4.7.6.2 (overvoltage) shall also be performed. The equipment shall then be examined for evidence of damage or deterioation.

4.8.7 Shock (See 3.14.2.7). - The test shall be conducted on the Shock Testing Machine for Light Weight Equipment shown in Specification MIL-S-901, fixture 4-C. The equipment shall be secured in its normal operating position to the test plate. The test shall consist of a total of 9 blows: one each 1 foot blow, 3 foot blow, and 5 foot blow, on the back, side, and top of the test plate with the blows being directed to the equipment respectively. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. During the test, the power supply shall be operated with a 230 VAC, 60 Hz, input under full load conditions (50 amps) at 28 VDC. After completion of the test, the unit shall remain operating (full load), for a period of not less than ten minutes. The unit shall then be examined for evidence of damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.8.8 Radio Frequency Interference (See 3.15). - Conformance of the sample with MIL-I-6181 shall be determined by the contractor or an approved testing laboratory. The contractor shall furnish the contracting officer, for approval, a test report, including details of the suppression system applicable for production units. The test report shall also include a block diagram of the test set-up and a list of the test equipment utilized. Upon approval by the contracting officer, and provided all other requirements of the specification are met, the approved test sample shall be used as a model for visual comparison of production units.

NOTE: When this test is performed under preproduction inspection the test results shall be included as part of the First Article Test Report.

4.8.9 Fungus test (See 3.7.2). - The power supply shall be subjected to 28 days of the fungus test specified in MIL-STD-810, Method 508 Procedure 1. After completion of the test, the power supply shall be connected to a single phase, 230 VAC, 60 Hz power source and operated under full output load conditions (28 VDC at 50 amperes) for a period of not less than ten minutes. The test item shall also be inspected for evidence of damage and deterioration.

4.9 Inspection for dimensional interchangeability. - The dimension listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.12. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug in assembly or major unit.

(e) Size and form of special threads.

4.10 Visual and mechanical inspection. - Equipment shall be examined for the defects listed in Standard MIL-STD-252 (See 3.19).

4.11 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with specification MIL-P-116 to determine conformance to the requirements of section 5.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

4.12 Production reliability testing shall be conducted as a sequential acceptance test in accordance with Test Plan IVA, using a specified MTBF of 7500 hours.

4.12.1 Measurements. - Measurements to be recorded shall not be taken prior to thirty (30) minutes of equipment "on" time. Each required performance parameter shall be measured and recorded at least once daily. If the value of any specified equipment parameter is not within the limits specified, a failure shall be recorded and the failure shall be presumed to have occurred immediately after the last successful measurement of the same parameter. All daily measurements shall be made at the nominal input voltage. The ten (10) minute vibration shall not be suspended during the time test measurements are taken. The sequence of selecting equipment for measurements shall be such as to assure the measurements for each parameter on each unit are taken at least once during the course of the test while under a 2.2G vibration. The nominal input voltage is 230 volts + 5, - 2%, 60Hz ± 5%. The following parameters shall be measured and recorded daily:

Parameter	Req. Para.	Insp. Para.
DC output current	3.11.2	4.7.1
DC output voltage	3.11.3	4.7.1

A failure is defined as the inability of the equipment to perform its required function. This includes catastrophic failures as well as any event which causes departure from performance as defined by this specification.

4.12.2 Equipment Quanitity. - The minimum number of equipments to be tested shall be ten (10), selected from each production lot.

4.12.3 Reliability Test Conditions. - The reliability test conditions shall be in accordance with Test Level A, paragraph 4.1.1 of MIL-STD-781.

4.12.4 Restoration of Equipment. - Upon completion of reliability tests all equipments shall be repaired by the contractor. Parts which have exceeded specified tolerance limits should be replaced and the repaired equipments shall be subjected to the normal acceptance procedures prior to shipment. Unless otherwise specified, the contractor shall furnish all replacement parts required as a result of the restoration.

4.12.5 Reports. - Monthly reliability progress reports, Final Test report and the Production Reliability Plan shall be furnished when required by the IFB and contract.

5. Preparation for Delivery

5.1 Preservation and Packaging. - Preservation and packaging shall be level A or C, as specified (See 6.2(b)).

5.1.1 Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

5.1.1.1 Cleaning. - Power supply, shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Power supply, shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit Packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116, as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged Method IC-1.

5.1.1.4.2 Power Supply PP-4763( )/GRC. - Each power supply shall be packaged method IA-14, as follows; Wind the A.C. cable on the mounting bracket and secure the plug end to the bracket with cotton tape or twine. Cushion the power supply on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, W5c, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned power supply within a close-fitting fiberboard box conforming to PPP-B-636, class domestic. Blunt all corners of the box. Place the boxed power supply within a barrier bag fabricated of water-vapor proof barrier material conforming to the requirements for class 1, application of MIL-B-131. Heat seal the closure. Place the bagged component within a second fiberboard box V3c, conforming to PPP-B-636. Close the box as specified in the appendix of the box specification.

5.1.2 Level C. - Power Supply PP-4763( )/GRC. Each power supply shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B or C as specified (See 6.2(b)).

5.2.1 Level A. -

5.2.1.1 Palletized load. - A quantity of containers packaged as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. - Power Supply PP-4763( )/GRC. When quantities per destination are less than a pallet load, the power supply packaged as specified 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

5.2.1.2 (Cont'd) Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B. -

5.2.2.1 Palletized load. - A quantity of power supplies packaged as specified in 5.1, shall be palletized as in 5.2.1.1.

5.2.2.2 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packaged as specified in 5.1, shall be packed within fiberboard boxes conforming to PPP-B-636, class domestic. Shipping containers shall be closed and reinforced as specified in the appendix of the box specification. Reinforcing shall be with pressure-sensitive filament tape conforming to PPP-T-97, type IV.

5.2.3 Level C. -

5.2.3.1 Palletized load. - A quantity of containers, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1, except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. - When quantities per destination are less than a pallet load, power supplies packaged as specified in 5.1, shall be packed as specified in 5.2.2.2.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. Notes

6.1 Intended use. - The power supply is used as a means of converting 115/230 volt ac, 50, 60 Hz single phase input sources to a nominal 28 volt dc, 50 amperes output suitable for operating Radio Set AN/GRC-106.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(c) The specific paragraphs of section 5 which are applicable to the particular procurement.

(d) Preproduction inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

(1) Three sample units of each item cited in section 1 are generally required so that lengthy and potentially damaging environmental tests can be performed on several sample units. (See 3.2)

(2) Detailed requirements covering preproduction pack(s), including inspection to be performed thereon and descriptive details, plans, drawings, etc. (Note: These may be obtained from the Packaging Branch, USAECOM, Philadelphia, Pa.

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required. (See 3.18)

(h) Quantity of tools and running spare parts required. (See 3.18)

(i) Submission of the statement of treatment referenced in 3.7.1, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.4 Nomenclature. - The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: PP-4763W/GRC. The contractor should apply for nomenclature (type designation) in accordance with the applicable clause in the contract. (See 1.1)

6.5 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.6 Qualitative material requirements. - The following service conditions were considered in the design of the equipment and are given here as a guide for the contractor where he is required to make a selection of parts, materials, processes, construction methods, etc.

6.6.1 Temperature. -

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the currentversion before use.

(a) Operating: Ambient temperature in the range of + 131°F to - 4°F. (The high temperature includes effect of sun-load). Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of + 160°F to - 80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

6.6.2 Relative humidity. - Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

6.6.3 Dust. - As encountered during heavy traffic over dry clay roads or in desert areas.

6.6.4 Sea salt fall-out. - Salt fall-out as experienced on the coast near warm bodies of water.

CUSTODIAN

Army-EL

PREPARING ACTIVITY

Army-EL

Project Nr. 5820-A491

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.